

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (E&C) (Credit System) Degree Examinations**

May - June 2023

20EC601 - ANTENNA AND MICROWAVE SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) The use of Smith chart is permitted.

Unit – I**Marks BT* CO* PO***

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|-------|--|----|-----|---|---|
| 1. a) | Explain the method of matching a transmission line using single stub matching technique quoting the general conditions at the appropriate steps. | 5 | L*2 | 1 | 1 |
| b) | The normalized admittance Y_r is given by $0.5 - j1.0$. Design a double stub matching system with $\lambda/2$ spacing. Also find the distance between the stubs. | 10 | L3 | 1 | 1 |
| c) | How the transformation of impedance is achieved using a quarter wave line? Explain. | 5 | L2 | 1 | 1 |
| 2. a) | With a neat construction diagram, explain a strip line. Write its characteristics. | 5 | L2 | 1 | 1 |
| b) | Explain the following terms for the microstrip lines.
(i) dielectric loss (ii) ohmic loss (iii) power handling | 8 | L2 | 1 | 1 |
| c) | A microstrip line uses a alumina substrate with relative permittivity of 9.7 and height of the slab 0.5 mm. The strip width is 0.5mm with effective dielectric constant of 6.556, characteristic impedance 50 ohm, free space wavelength 1.5 cm. Assume $\tan\delta = 2 \times 10^{-4}$ and $t = 0.2$ mm. If $f = 4$ GHz, calculate the total attenuation (assume copper with conductivity of 5.7×10^7). | 7 | L3 | 1 | 1 |

Unit – II

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|-------|---|----|----|---|---|
| 3. a) | With a neat construction diagram, derive the S matrix for an E-plane Tee junction. Use the properties of S matrix. | 10 | L2 | 2 | 1 |
| b) | Explain the construction and features of EH – plane Tee junction. Derive the S matrix. | 10 | L2 | 2 | 1 |
| 4. a) | With a neat construction diagram and applegate diagram, explain the operation a Reflex klystron. | 10 | L2 | 2 | 1 |
| b) | Explain the Gunn mode and Limited space charge accumulation modes of operations of a Gunn diode with the necessary figures. | 10 | L2 | 2 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 5. a) | Explain the following parameters of antenna system.
(i) beam area
(ii) normalized E-field
(iii) normalized power
(iv) radiation intensity | 8 | L2 | 3 | 1 |
| b) | Define effective aperture. Derive an expression for effective aperture in terms of directivity. | 7 | L2 | 3 | 1 |
| c) | Determine the directivity of an antenna whose radiation resistance is 72Ω . The loss resistance is 12Ω and gain is 20. | 5 | L3 | 3 | 1 |
| 6. a) | Derive an expression for total field in the case of two isotropic point sources of equal amplitude and opposite phase. Draw the pattern and vector diagram. | 8 | L2 | 3 | 1 |

P.T.O.

- b) Determine the directivity for the following pattern,

$$U = U_m \sin \theta \sin^2 \phi; 0 \leq \theta \leq \pi$$

$$0 \leq \phi \leq \pi$$

7 L3 3 1

- c) Calculate the approximate directivity from the half-power beam widths of a unidirectional antenna if the normalized power pattern is given by $P_n = \cos \Theta$.

5 L3 3 1

Unit – IV

7. a) Derive expressions for null direction finding for an array system. How it can be applied to broadside and end-fire array system?
 b) For a short dipole placed in x, y and z coordinate system, derive an expression for retarded vector potential in the case of a short dipole.
8. a) Derive an expression for radiation resistance of a short dipole antenna.
 b) Derive expression for E and H field intensities for a square loop antenna placed in x, y and z coordinate system.

10 L2 4 1

10 L2 4 1

10 L2 4 1

10 L2 4 1

Unit – V

9. a) Derive an expression for relative phase velocity for a monofilar helical antenna in mode.
 b) Explain the construction and design of a Yagi-uda antenna.
 c) Distinguish between center feed and off-center feed types in slot-antenna.
10. a) Explain the transmission modes of monofilar helical antenna.
 b) Discuss the essential features and design equations of a horn antenna.
 c) State and explain Babinet's principle of optics. Also explain how the principle can be incorporated in dipole antenna.

10 L2 4 1

5 L2 4 1

5 L2 4 1

5 L2 4 1

5 L2 4 1

10 L2 4 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
